Fiona: Project Documentation

Oct 11, 2026

Fiona is the AI finance council for small businesses: it advises like a CFO, acts only within the mandate the owner gives it, and nothing moves until an independent Guardian has checked it.

FinTechathon 2026 · International AI Track · Direction E: SME Finance Copilot · Theme: *AI as a Financial Participant*

Team member	Role
Abdullah	Finance strategy and decision intelligence
Asharib Khan	Architecture and AI engineering
Daniyal	Scale and personalization
Yahya	Forecasting and market intelligence

Executive summary

Fiona gives a small business the finance team it cannot afford to hire: a council of AI agents that reads its numbers, recommends decisions and carries out routine money tasks within limits the owner sets. Every action passes an independent Guardian before it runs, and every step is recorded.

What it does

- Works out which life-cycle stage the business is in from its own cash flows, then advises on borrowing versus raising equity, investing, paying dividends, cutting debt and launching products.
- Reconciles bank, point-of-sale and invoice data, catches duplicate and suspicious bills, and verifies customer and supplier invoices.
- Forecasts cash 13 weeks ahead, schedules payments, and pays known suppliers from a policy-bound agent wallet.
- Tests a new product idea against demand signals, unit economics and a probabilistic forecast before money is spent.

- Reads statements prepared under IFRS, IFRS for SMEs, US GAAP or China's Accounting Standards for Business Enterprises (CAS) and compares them on one basis.

What is new. Incumbents are adding AI agents that draft entries inside one ledger. Fiona's difference is structural: authority is delegated per agent and per action type, an independent Guardian checks every action, advice adapts to the firm's life-cycle stage, and the ledger is framework-neutral. The combination turns the AI from an assistant into an accountable participant, which is what AI as a Financial Participant asks for.

What the demo proves. One five-minute story: a business onboards, gets its morning briefing, watches the Guardian block a fraudulent bill, sees the agents debate a loan against equity, simulates a product launch, approves a payment by phone, and then the same engine reconfigures itself for a 300-person distributor in Shenzhen.

The problem

Small businesses are most of the economy, yet they make capital decisions without a CFO and with data scattered across a dozen places.

- MSMEs are over 90% of firms and about 70% of employment worldwide (IFC).
- Across APEC economies, MSMEs are more than 90% of businesses and employ over half the workforce (APEC).
- In 119 emerging and developing economies, the MSME finance gap is about US$5.7 trillion, 19% of their GDP (World Bank, March 2025 report).

Gap	What happens today	What it costs
No finance leadership	Owners decide to borrow, raise equity, reinvest or launch on instinct	Expensive debt, idle cash, launches that drain working capital
Scattered data	Bank statements, POS exports, bills on WhatsApp and spreadsheets never meet	Manual reconciliation; errors and duplicate payments slip through
Weak controls	One person often orders, receives and pays	Changed supplier bank details, duplicate bills and refund abuse go unnoticed

Gap	What happens today	What it costs
Framework complexity	Statutory, tax and lender reports follow different rules	Owners cannot compare or explain their own numbers
Invisible to lenders	Banks cannot underwrite what they cannot see	Credit is refused or priced high, and the gap persists

The framework load is about to grow. IFRS 18, which replaces IAS 1 and reshapes the income statement, takes effect for periods beginning 1 January 2027 (IFRS Foundation), and so does the third edition of the IFRS for SMEs Standard (Forvis Mazars).

The 2026 competition puts AI-driven inclusive finance at its core (CLS, 2 Sep 2026). Inclusive finance starts with data: a business whose books are clean, reconciled and explainable is a business a bank can lend to.

Our thesis: AI as a Financial Participant

An AI becomes a financial participant, not a tool, when it holds what a human finance employee holds: a role, a mandate, a check on its work, and a record of what it did. Fiona is built around those four.

Element	In Fiona	Human equivalent
Role	Each agent owns one finance job: strategy, treasury, control or growth	Job description
Mandate	Limits the owner sets on what each agent may do alone: amounts, payees, action types	Signing authority
Guardian	An independent checker reviews every proposed action before it runs	Maker-checker (four-eyes) control
Record	Every action keeps its evidence, reasoning and approvals in a tamper-evident log	Audit trail

The autonomy ladder. The owner decides how far each agent may go, per type of action:

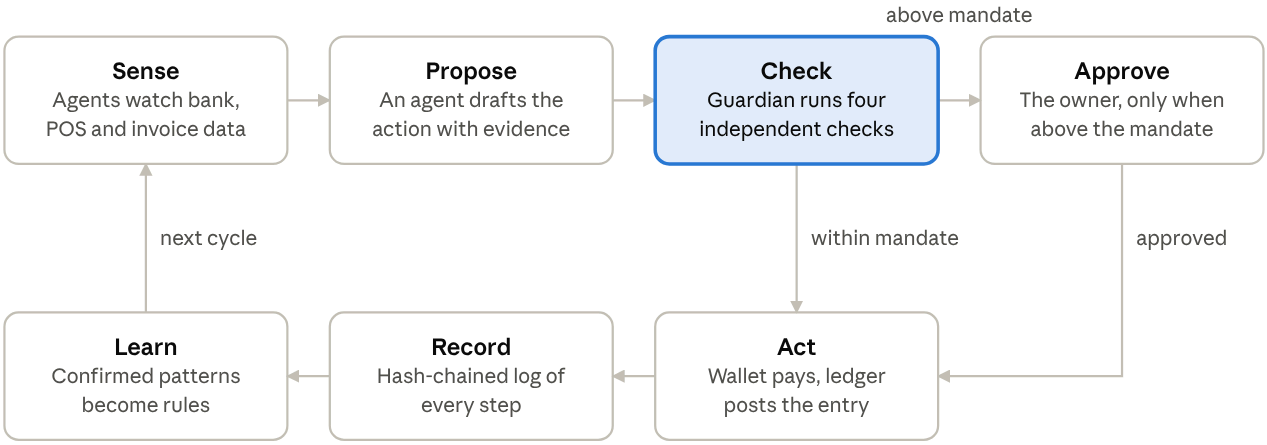
1. **Advise:** the agent recommends; the owner acts.
2. **Prepare:** the agent drafts the invoice, payment or journal entry; the owner approves with one tap.

3. **Act within mandate:** the agent executes inside its limits, the Guardian checks first, and the owner gets a daily digest.

Every agent starts at Advise. Trust is earned action by action, and the owner can move any agent back down at any time.

The agent wallet. This is where the track's *Agent Wallet* theme meets Direction E. Fiona's Treasurer holds a ring-fenced business wallet with programmable policies: per-payment and daily caps, an allow-list of payees, allowed categories and payment windows. It can pay a known supplier's due bill on its own; a new payee, a changed bank account or anything over the cap stops for the Guardian and the owner.

Nothing runs until the Guardian has checked it



participation loop · 7 steps, 2 mandate paths

Inside its mandate, an agent's action goes straight from the Guardian to execution; above the mandate, the owner approves first.

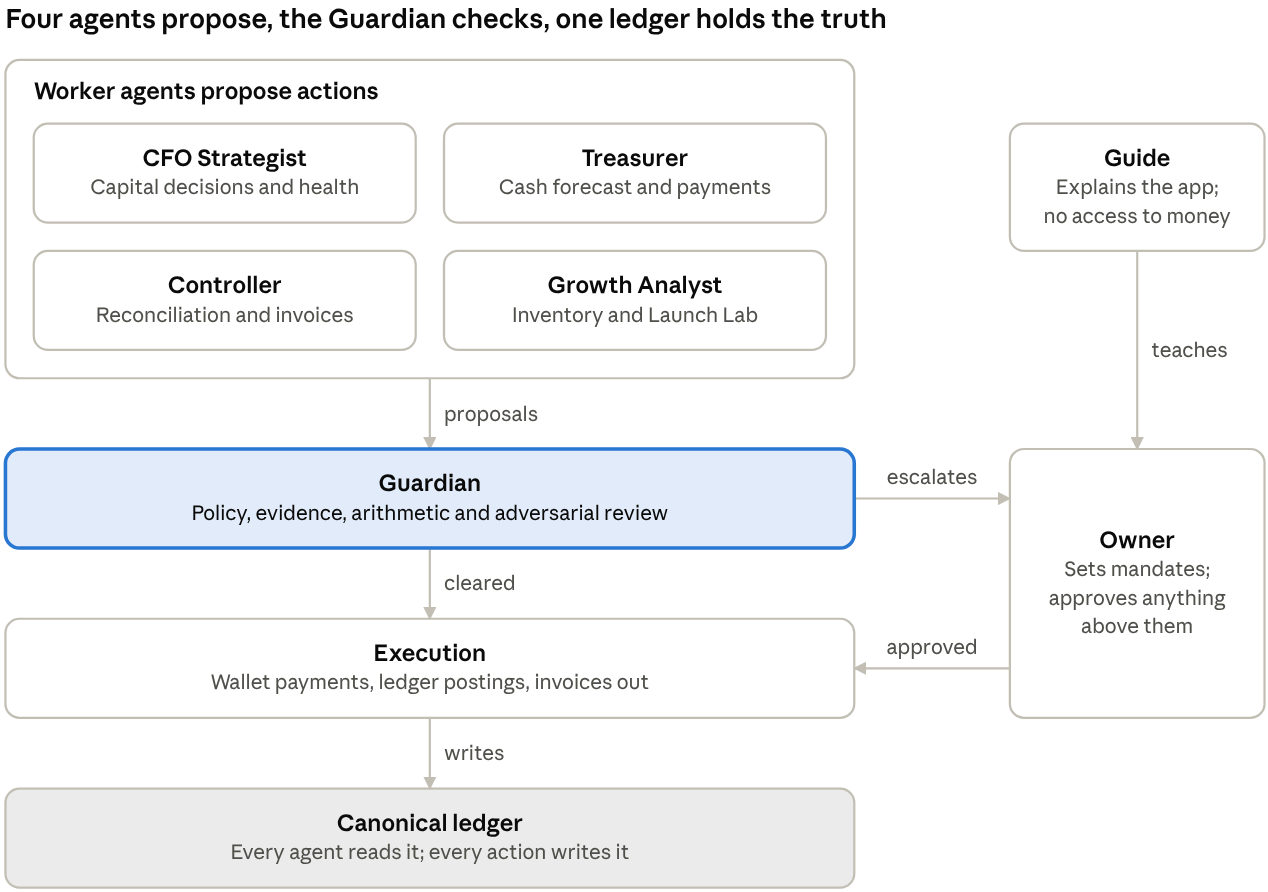
Product overview: the AI finance council

Fiona is six agents around one ledger: four do the work, one checks it, and one teaches the product.

Agent	Job	Typical actions	Highest autonomy allowed
CFO Strategist	Capital decisions and business health	Health score, stage detection, debt versus equity, dividends versus reinvestment, capex, debt reduction	Advise
Treasurer	Cash and payments	13-week cash forecast, payment scheduling, early-payment discounts, agent-wallet payments	Act within mandate
Controller	Books and controls	Bank, POS and invoice reconciliation, duplicate detection, customer invoices, supplier bill checks, payment orders	Prepare
Growth Analyst	Products and inventory	Fast and slow movers, reorder points, dead stock, Launch Lab simulations	Advise
Guardian	Independent check	Policy rules, second-model review, recomputed figures, evidence check	Can block, never acts
Guide	Help with the product	Guided tours, "what does this number mean?", feature help	No access to money

One ledger, many lenses. Every agent reads the same canonical ledger, so the Treasurer's cash view and the CFO's balance sheet can never disagree. The Guardian sees only the proposal and the raw data, never the proposing agent's reasoning, so its check is independent by construction.

Council mode. For decisions that change the business, such as a loan, a new branch or a dividend, the agents meet as a board. The CFO proposes, the Treasurer tests the cash impact, the Growth Analyst argues the upside and the Guardian argues against. The owner receives one decision memo with the recommendation, the numbers behind it and the dissent on record.



agent map · 6 agents, 1 ledger

The worker agents never reach execution directly: every proposal passes the Guardian, and anything above the mandate goes to the owner.

Core capabilities

Eight capabilities carry the product. The first four run the money; the last four grow it, guard it and explain it.

1. Business profile and Stage Detector

Onboarding asks for industry, size, legal form, country and reporting framework. Fiona then infers the life-cycle stage from the signs of operating, investing and financing cash flows, the method of Dickinson (2011), *The Accounting Review* 86(6). When the owner's own description and the cash flows disagree, Fiona says so and explains why.

Stage	Operating	Investing	Financing	What Fiona prioritises
Introduction	Out	Out	In	Runway, burn rate, grants and equity
Growth	In	Out	In	Working capital, growth financing mix, capacity
Mature	In	Out	Out	Dividends versus reinvestment, cost of debt, efficiency
Shake-out	Any other pattern			Margins, cost structure, focus
Decline	Out	In	In or out	Liquidity, restructuring, asset sales

2. Decision engine

The CFO Strategist turns the statements into five recurring decisions. Every figure is computed by the finance engine and explained by the model, never estimated by the model.

Decision	What Fiona checks	What the owner gets
Debt or equity	Leverage, interest cover, debt-service coverage, collateral, cost of debt versus equity, dilution, stage	Recommended mix, with coverage and runway under each option
Dividend or reinvest	Return on invested capital versus cost of capital, cash buffer, investment pipeline, distributable reserves	A payout range and the return reinvestment would need to earn
Invest in assets	NPV, IRR, payback, lease versus buy, capacity use	Go or no-go, with break-even and sensitivity
Reduce liabilities	Rate per facility, prepayment terms, working capital that can be released	Pay-down order and cash freed per month

Decision	What Fiona checks	What the owner gets
Innovate or adopt technology	Free cash flow after maintenance spending, stage, payback horizon	An innovation budget and a pilot size

Each recommendation shows its working: the ratios, the thresholds and the source rows. If the data is too thin to decide, Fiona says what it needs instead of guessing.

3. Treasury

- **13-week cash forecast** from open invoices, bills, payroll, loan schedules and sales trends, flagging the first week cash falls below the owner's buffer.
- **Payment timing:** bills are paid on their due date unless an early-payment discount beats the cost of money; a 2/10 net 30 discount is worth about 37% a year.
- **Agent wallet:** routine payments to known payees inside the mandate; everything else goes to the Guardian and the owner.
- **Credit-ready pack:** reconciled statements, ratios and the cash forecast, shared with a lender only with the owner's consent.

4. Controller

- **Reconciliation in four passes:** exact matches, then rule matches within amount and date tolerances, then one-to-many matches (one deposit covering several settlements), then model-assisted matches for what is left, each with a confidence score. Confirmed matches become rules.
- **POS reconciliation:** register totals against card, wallet and bank settlements, cash over or short per drawer, and void and refund rates per cashier.
- **Duplicate detection:** same supplier, an amount within tolerance, near-identical invoice numbers (INV-1001 and INV1001), or the same file.
- **Customer invoices:** issued with tax fields and a payment link or QR code, matched to the bank line on payment, and followed up politely, then firmly.
- **Supplier bills:** read from PDF, photo or email, then checked against the purchase order and goods received (three-way match), tax registration, arithmetic, price history and any change in bank details. Where a tax authority makes invoices verifiable, as with FBR-integrated POS invoices in Pakistan, Fiona checks that too.
- **Payment orders:** bank payment files or pay-order requests, released only after Guardian clearance and approval.

5. Inventory intelligence

- **ABC–XYZ classification:** A, B or C by value contribution and X, Y or Z by demand variability, so the owner knows which items deserve tight control.
- **Fast movers, slow movers and dead stock** from POS line items, with the cash each one ties up.
- **Reorder points:** expected demand over the supplier's lead time plus safety stock, with suggested quantities and draft purchase orders.
- **Margin per item** and gross margin return on inventory, including size-and-colour variants and bundles.

6. Launch Lab

Before an owner commits money to a new product, Launch Lab answers three questions: will it sell, will it pay back, and can the business afford it.

1. **Inputs:** investment in stock, equipment and marketing, price, unit cost, supplier lead time and launch date.
2. **Demand signals:** search-interest trends, marketplace listings and prices, the firm's own sales of similar items, and seasonal peaks such as Ramadan, Eid or Chinese New Year.
3. **Forecast:** demand curves borrowed from similar products, plus the open-source Chronos-2 model, which forecasts without training on the firm's data; a Monte Carlo run then produces low, base and high cases.
4. **Economics:** break-even units and month, payback, NPV, and how much the new item would take from existing ones.
5. **Affordability:** the Treasurer runs the cash forecast with the launch in it, and the Guardian checks the cash buffer.

The output is a launch memo: go, pilot first, or don't, with the probability of breaking even within the owner's horizon. A single shop's demand is rarely observable directly, so Launch Lab works from proxies and shows how confident it is.

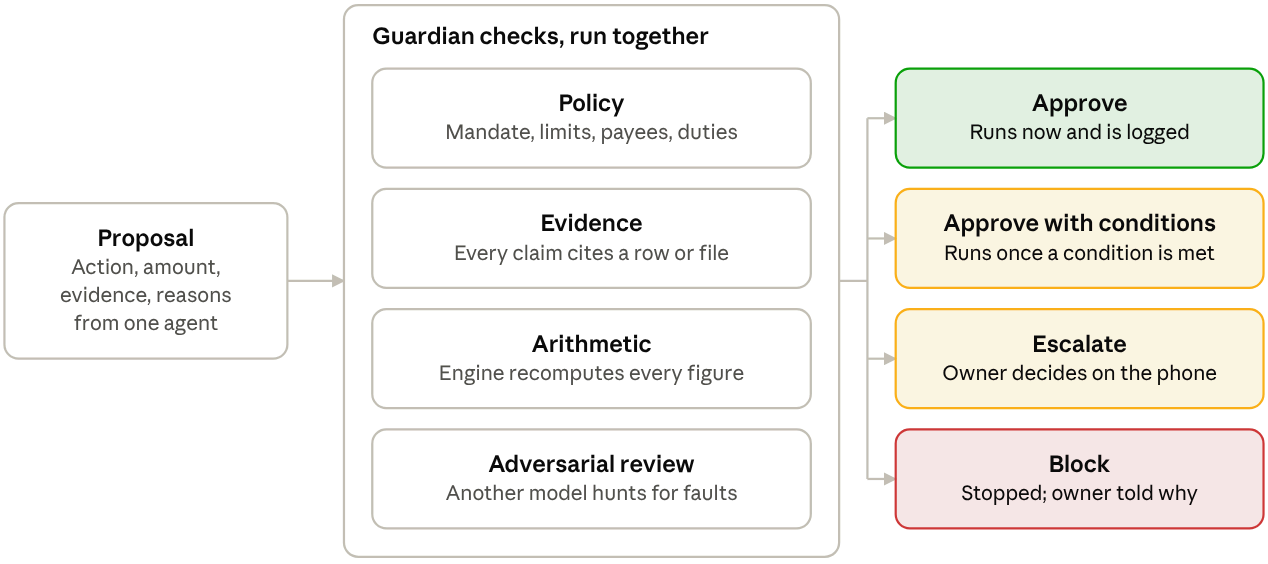
7. The Guardian

The Guardian is the second key: no money moves, no invoice goes out and no entry posts until it has checked the action.

Check	How it works	What it catches
Policy	Deterministic rules from the mandate and internal controls	A payment over the cap, a new payee, a recently changed bank account, the same person raising and approving
Evidence	Every claim in a proposal must cite a ledger row or a document	A bill with no purchase order behind it
Arithmetic	The finance engine recomputes every figure	Lines that do not add up, wrong tax
Adversarial review	A different model, asked to find reasons the action is wrong, sees the proposal and raw data but not the proposer's reasoning	An "urgent" request to pay a new account

Each check ends in approve, approve with conditions, escalate to the owner, or block, with the reasons stored. Bills and emails are treated as data, never as instructions: a message saying "pay this now" is evidence to examine, not a command. Every record carries the hash of the one before it, so any edit to history breaks the chain.

Every proposal passes four checks and leaves with one of four outcomes



Guardian flow · 4 checks, 4 outcomes

All four checks run on every proposal, and one failure is enough to escalate or block it.

8. The Guide

The Guide is the assistant over the assistant. It lives in a side panel, knows the product, the current screen and the user's role, and answers "how do I…?" and "what does this number mean?" with guided tours that highlight the right control. It is kept apart from the finance agents on purpose: it explains the product, never gives financial advice, and has no access to money. It also records where users get stuck, so the team knows which screens to fix.

Multi-framework engine

There is no single financial reporting framework; there are several, and Fiona treats them as lenses on one set of facts. The useful question is not "which framework?" but "which lens, for which reader?"

Framework	Set by	Who uses it
IFRS Accounting Standards	IASB	Listed and public-interest companies in most of the 170 jurisdictions the IFRS Foundation profiles
IFRS for SMEs	IASB	Private SMEs where adopted; third edition from 1 January 2027
US GAAP	FASB	Companies reporting in the United States
CAS (Accounting Standards for Business Enterprises)	Ministry of Finance, PRC	Companies in mainland China; substantially converged with IFRS
National small-entity standards	National standard setters	Pakistan, for example: IFRS for listed and large companies, IFRS for SMEs for medium ones, Revised AFRS for SSEs for small ones (IFRS Foundation, Pakistan profile)
Cash or tax basis	Tax law	The only books many micro businesses keep

How Fiona handles every framework

1. **Canonical facts.** Each transaction is stored once, at the finest grain, in framework-neutral form: "sold 3 units for 9,000 on credit; cost 5,400".
2. **Rulepacks.** Each framework is a versioned package of recognition, measurement and presentation rules plus a chart-of-accounts mapping. The same facts run through any rulepack to produce that framework's statements.
3. **Taxonomy mapping.** Line items map to the IFRS and US GAAP XBRL taxonomies, so Fiona can read published filings and export structured statements.
4. **Reading the unfamiliar.** When a user uploads statements in any framework or language, Claude identifies the framework from the basis-of-preparation note and maps each line to canonical concepts with a confidence score. The engine then checks that everything still balances, and low-confidence lines go to a person.

Differences the rulepacks encode, for example

Topic	IFRS	US GAAP	Why it matters
Inventory cost formula	LIFO not allowed	LIFO allowed	Cost of sales and tax under inflation
Inventory write-down	Reversed when value recovers	Not reversed	Margins after a markdown
Property, plant and equipment	Revaluation model allowed	Cost model only	Equity and leverage
Development costs	Capitalised when criteria are met	Generally expensed	Profit of product companies
Leases, lessee	One on-balance-sheet model	Operating and finance leases	EBITDA and leverage ratios

Honest scope. The hackathon build ships rulepacks for IFRS, IFRS for SMEs and US GAAP on these topics, plus CAS presentation mapping. New rulepacks plug in without changing the engine. Audit-ready statutory statements in every framework come later and need qualified accountants to review each rulepack.

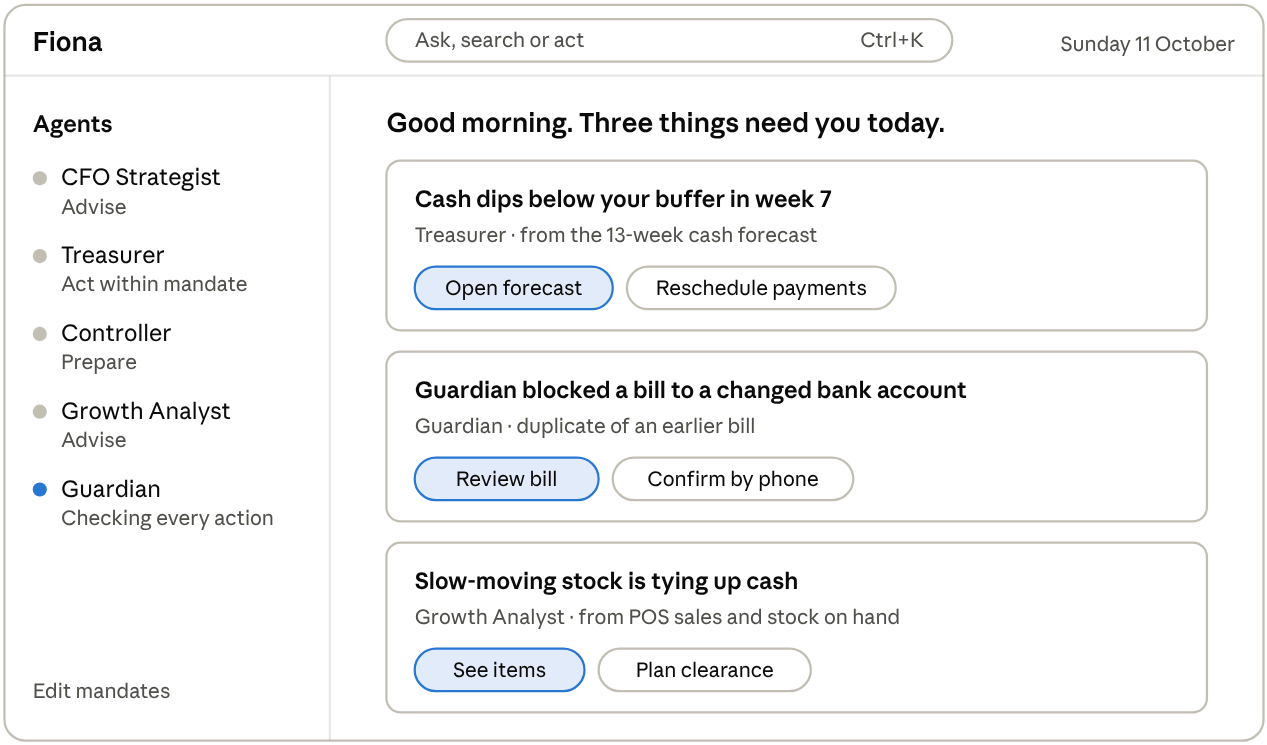
Experience design

Fiona opens on a briefing, not a chat box. The owner sees what changed overnight and what needs a decision, and every card can be acted on in one tap.

- **Briefing first.** The agents write the home screen each morning: three to five cards ranked by money at stake, such as "cash dips below your buffer in week 7" or "the Guardian blocked a bill sent to a changed bank account".
- **Generative interface.** Agents answer with live components streamed into the page, such as a chart, a comparison table or an approval card, instead of paragraphs of text.
- **Explain any number.** Every figure opens its definition, formula, source rows and the agent that produced it.
- **Command bar.** One shortcut (Ctrl+K) to ask, search or act, typed or spoken, in English, Urdu or Chinese.
- **The council table.** Council mode lays each agent's position side by side, marks where they disagree, and ends with the memo.
- **Trust made visible.** Each agent shows its autonomy level, its mandate and its recent actions; one switch moves it up or down the ladder.
- **Mobile companion.** Biometric approvals, alerts, receipt capture by camera and the briefing. It is a companion, not a shrunken dashboard.

Visual direction. Calm light and dark themes, one accent colour, and numbers in a tabular typeface so columns line up. Motion appears only to show change: a figure counting to its new value or a card arriving. "Futuristic but friendly" means fewer screens, not more effects.

The home screen is a briefing the agents write, not a chat box



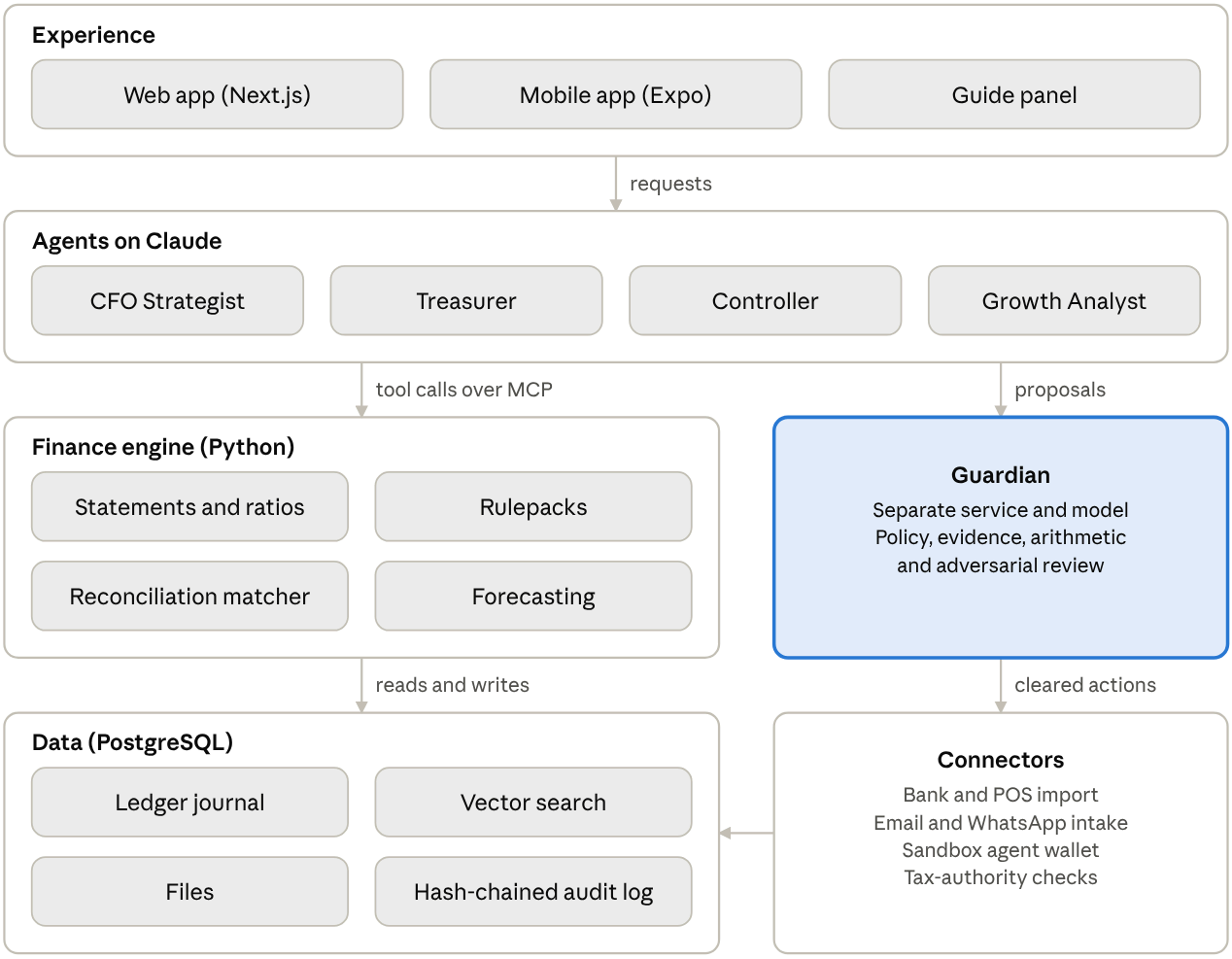
home screen mock-up · briefing, agent rail, command bar

The owner starts from three ranked cards with one-tap actions; the agent rail shows how far each agent is trusted to act.

System architecture

Fiona has five layers and one rule that runs through all of them: the model decides what to compute and explains the result, deterministic code does the computing, and the Guardian checks before anything executes.

The model reasons, the engine computes, the Guardian checks



system architecture · 5 layers plus the Guardian

Agents reach the engine for analysis but reach the outside world only through the Guardian.

Layer	Components	Responsibility
Experience	Next.js web app, Expo mobile app, the Guide	Briefing, council view, approvals, explanations
Agents	CFO Strategist, Treasurer, Controller, Growth Analyst; the Guardian as a separate service	Plan, propose, explain; check independently

Layer	Components	Responsibility
Finance engine	Python service: statements, ratios, rulepacks, reconciliation matcher, forecasting, Monte Carlo, decision models	Every number the product shows
Data	PostgreSQL ledger with an append-only journal, pgvector, object storage, job queue, hash-chained audit log	One source of truth and a tamper-evident record
Connectors	Bank statement and POS import, email and WhatsApp intake, sandbox wallet, tax-authority checks, accounting-system import	Getting data in and actions out

One request, end to end: paying a supplier's bill

1. The Controller reads the bill with Claude and extracts its fields into a fixed schema.
2. The finance engine runs the three-way match and the duplicate check.
3. The Treasurer picks the payment date from the cash forecast and drafts the payment.
4. The Guardian runs its four checks.
5. Inside the mandate, the wallet pays; outside it, the owner approves on the phone.
6. The ledger posts the entry and the audit log appends a hash-chained record.

Work	Done by
Reading bills and statements, mapping line items, explaining results, drafting memos, debating options	Claude
Ratios, statements, matching, forecasts, NPV, tax, policy checks	Finance engine (code)
The final word on money	The owner, within the mandate

Technology stack

Fiona is one platform delivered three ways: a web application, a mobile companion app and an API. The stack keeps the team in two languages: TypeScript for everything a user touches, Python for everything that computes.

Layer	Choice	Why
Web	Next.js 16 (App Router) on React 19.2, TypeScript	Server rendering for fast dashboards; streaming for live agent output
UI	Tailwind CSS v4, shadcn/ui on Radix, Motion, Apache ECharts, TanStack Table and Query	Accessible components we own; charts that handle large series
Agent streaming in the UI	Vercel AI SDK with the Anthropic provider	Streams tool results as React components (generative UI)
Mobile	Expo SDK 56 (React Native 0.85, React 19.2), Expo Router	Same language and React model as the web; biometrics, camera, push
Monorepo	Turborepo with pnpm workspaces	Shared types, Zod schemas, API client and design tokens
Finance engine	Python 3.12+, FastAPI, Pydantic v2, Polars, statsmodels, StatsForecast, Chronos-2, scikit-learn, RapidFuzz, Arelle (XBRL)	The strongest ecosystem for finance math, forecasting and matching
Agents	Claude API through the Anthropic Python SDK; the finance engine exposed as an MCP server	One tool interface for every agent, reusable by other MCP clients
Data	PostgreSQL 17+ on Supabase with row-level security, pgvector, Supabase Storage and Auth	Ledger, search, files and login in one managed service
Jobs	Arq on Redis	Nightly runs, forecasts, reminders

Layer	Choice	Why
Observability	Sentry, PostHog, Langfuse	Errors, product analytics, model traces and evaluations
Deployment	Vercel (web), Fly.io (engine), Supabase (data), EAS (mobile builds)	Free or low-cost tiers for the competition
Testing	Vitest and Playwright; pytest with Hypothesis	Property tests such as "debits always equal credits"

On "React 19 for the phone". React 19 is the UI library that Next.js already runs on; it does not build phone apps by itself. React Native does, through Expo, and it runs React 19.2 too, so the web and the app share one language, one mental model and one set of types. Flutter is the alternative if someone on the team is much faster in it, at the cost of that sharing. If time runs short, a progressive web app from the Next.js build is the fallback.

What we chose not to use. A single TypeScript monolith would force finance math and forecasting into a weaker ecosystem. A blockchain for the audit log would add operational weight for little gain; a hash-chained log already makes tampering evident, and anchoring its latest hash to an external ledger such as FISCO BCOS is a later option for multi-party audit.

AI design with Claude

Claude does the reading, reasoning and explaining. Each agent uses the smallest model that does its job well, and the Guardian always runs on a different model from the agent it checks.

Role	Model (overview)	Why
Bill and statement extraction, categorisation, the Guide	Claude Haiku 5.5 (`claude-haiku-5-5`)	High volume and low latency
CFO Strategist, Treasurer, Growth Analyst, council debate	Claude Sonnet 5.5 (`claude-sonnet-5-5`)	Strong reasoning at a sustainable cost

Role	Model (overview)	Why
Guardian adversarial review, final memo on large decisions	Claude Opus 5.5 (`claude-opus-5-5`)	The strongest reviewer where mistakes cost most, and a different model from the maker

Claude API feature	Use in Fiona
Tool use with strict schemas	Agents call finance-engine tools; every proposal is a typed object
Structured outputs	Extractions and proposals validated against JSON schemas
PDF support and vision	Bills, receipts, bank statements and scanned annual reports
Citations	Each claim in a memo links to the page or row it came from
Prompt caching	Company profile, chart of accounts and rulepacks reused across calls
Batch processing	Overnight categorisation and re-reconciliation at lower cost
Memory tool	The owner's preferences and past decisions carried across sessions
Managed Agents permission policies (later)	Hosted, scheduled agents with enforced permissions and budgets

Guardrails

- Numbers never come from the model's head: tools compute them and the model cites the tool result.
- Bills, emails and uploaded files enter as data; instructions inside them are never followed.
- Each agent holds only the tools its role needs, and the Guide holds none that touch money.
- Anything outside the mandate needs the owner, with biometric approval on the phone.

- When data is thin, agents ask for what they need rather than guess.

How we will know it works

- **Planted-issue tests:** the synthetic dataset hides duplicates, changed bank details, wrong tax and a cash crunch; we measure how many the Guardian and Controller catch and how many false alarms they raise.
- **Reconciliation accuracy** against the generator's ground truth, which knows every correct match.
- **Framework mapping accuracy** on published filings: US GAAP statements from SEC EDGAR and IFRS statements from filings.xbrl.org.
- **Decision-memo review** by Abdullah against a rubric: correct ratios, correct direction, assumptions stated.
- **Regression evaluations** in Langfuse on every prompt or model change.

Data: what Fiona needs and where the demo data comes from

Fiona needs seven inputs to run its core features. The demo runs on a synthetic business built so that every planted problem has a known answer, with public datasets for forecasting and framework tests.

Input	Format	Key fields	Feeds
Company profile	Onboarding form	Industry, size, legal form, country, currency, framework, fiscal year	Personalisation, rulepack choice
Bank statements, 12 to 24 months	CSV or PDF	Date, narration, amount, balance, reference	Reconciliation, cash forecast, Stage Detector
Sales (POS) transactions	CSV export	Time, item, variant, quantity, price, discount, payment method, cashier	Inventory, margins, POS reconciliation
Customer invoices and supplier bills	CSV, PDF or photo	Party, number, dates, lines, tax, status	Receivables, payables, duplicates, three-way match

Input	Format	Key fields	Feeds
Inventory and suppliers	CSV	Item, unit cost, stock on hand, reorder level, supplier, lead time	Reorder points, Launch Lab
Last year's statements or opening balances	Excel or PDF	Balance sheet, income statement, cash flow statement	Decision engine, framework mapping
Loans and obligations	Form	Lender, principal, rate, schedule, covenants	Debt decisions, cash forecast

Optional inputs deepen the analysis: payroll summary, fixed-asset register, budget, purchase orders, goods-received notes and tax filings.

Every record lands in the ledger as a framework-neutral event:

```
{"event": "sale", "date": "2026-10-03", "party": "C-0142",
 "lines": [{"item": "KRT-M-BLU", "qty": 3, "price": 3000, "cost": 1800}],
 "payment": {"method": "bank_qr", "status": "settled"},
 "source": {"type": "pos_export", "row": 4412}}
```

Demo data plan

1. **Synthetic business generator.** A Python generator creates 24 months of a coherent business in which bank lines really match invoices and POS settlements. It plants known problems: a duplicate supplier bill, a changed supplier bank account, a refund spike at one register, slow-moving stock, a seasonal peak and a cash dip a few weeks ahead. It also writes the answer key, so every demo moment and every accuracy test can be checked.
2. **Two personas.** A Karachi apparel retailer reporting under IFRS for SMEs in PKR, and a 300-person Shenzhen electronics distributor reporting under CAS in CNY. Switching between them shows personalisation, scale and framework handling.
3. **Public datasets** for the parts synthetic data cannot prove: UCI Online Retail II for basket and demand patterns, the Kaggle Favorita store-sales and M5 datasets for forecasting, SEC EDGAR financial statement data sets for US GAAP statements, filings.xbrl.org for IFRS statements, and Aswath Damodaran's datasets for industry betas and country risk premiums.
4. **One real business, if possible.** Even three months of anonymised data from a real shop makes one demo moment unforgettable; judges remember real numbers.

Real data is anonymised before use, only the fields above are stored, and nothing is shared with a lender without the owner's consent.

Scale and personalization

The same engine serves a single shop and a multi-entity group. Three dials set the experience: the life-cycle stage, the industry pack and the size tier. The code does not change.

Tier	Typical business	What switches on
Micro	One owner, one location, cash or tax-basis books	Briefing, photo and WhatsApp capture, simple cash forecast, Guide-led setup
Small	A few staff and locations, spreadsheets or basic software	Full reconciliation, receivables and payables, agent wallet, Launch Lab
Medium	Hundreds of staff and a finance team	Roles, approval chains, segregation of duties, budgets versus actuals, multiple locations
Large or group	Several legal entities on an ERP	Consolidation, inter-company eliminations, multiple currencies, ERP connectors, single sign-on, data residency

Industry pack	Extra measures	Extra workflows
Retail and fashion	Sell-through, markdown rate, gross margin return on inventory, size and colour mix	Seasonal buying plans, consignment
Food and beverage	Food-cost percentage, prime cost, waste	Recipe costing: selling a burger deducts the bun, patty and sauce
Manufacturing	Capacity use, yield, standard-cost variances	Bills of materials, work in progress
Services	Utilisation, realisation, days sales outstanding	Time-based billing, retainers

Industry pack	Extra measures	Extra workflows
Wholesale and distribution	Fill rate, days of inventory, credit exposure per customer	Customer credit limits, route settlements

How it scales technically

- Multi-tenant PostgreSQL with row-level security; the largest tenants move to dedicated databases.
- The finance engine runs as stateless containers that scale out, with heavy jobs on queues.
- Above a data-size threshold, analytics move to a column store while the ledger stays in PostgreSQL.
- ERP connectors (SAP, Oracle, Odoo, Kingdee) sit behind one integration layer, so agents see the same canonical events whatever the source.

Security, compliance and trust

A participant that can move money must be safer than the person it helps. Fiona follows the principles banks already use: least privilege, segregation of duties, maker-checker and a complete audit trail.

Control	How Fiona applies it
Encryption	TLS in transit, encryption at rest for the database and files, secrets in a managed vault
Access	Role-based permissions, two-factor login for owners and admins, biometric approval on mobile
Segregation of duties	No agent or user both proposes and approves the same payment
Payment safety	Allow-listed payees, caps, a cooling-off period after any bank-detail change, out-of-band confirmation for new accounts
Audit	A hash-chained record of every proposal, check, approval and execution

Control	How Fiona applies it
Data minimisation	Only the fields Fiona needs are stored; demos and evaluations use synthetic or anonymised data
Model data	Claude API under commercial terms, with data residency options where available
Card data	Never stored by Fiona; payment partners hold it

Regulatory posture

- **Decision support, not regulated advice.** Fiona recommends and shows its working; the owner decides. Investment ideas appear as options with their risks.
- **Licensed partners move the money.** The agent wallet is a sub-account or virtual card issued by a licensed bank or payment institution; Fiona never holds client funds.
- **Local tax rules as versioned packs.** Invoice formats and e-invoicing hooks, such as FBR-integrated POS invoices in Pakistan and China's digital fapiao, live in compliance packs that are reviewed whenever the rules change.
- **Personal data.** Consent, purpose limitation and deletion on request by default, with jurisdiction rules applied on top, such as China's Personal Information Protection Law for the Shenzhen persona.

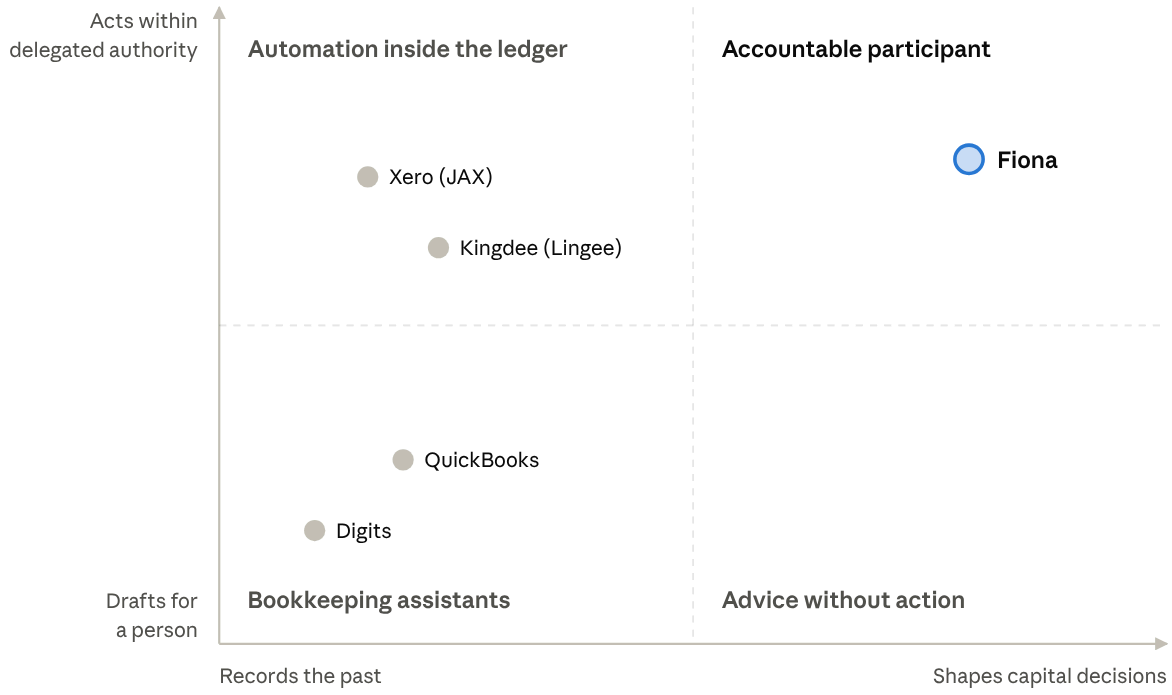
Competitive landscape

The incumbents are moving fast into agents, so Fiona does not compete on features they already have. It competes on how the AI participates: delegated authority, independent checking, stage-aware capital advice and framework neutrality. Analysis as of 11 October 2026.

Competitor	What they have now	Where they are strong	Gap Fiona targets
Intuit QuickBooks	A team of AI agents for accounting, finance, payments, customers and projects since July 2025, later joined by tax and payroll agents (overview)	The largest SMB base in the US, payroll and tax depth	Agents draft for approval; the deepest agents sit on higher-priced plans; one framework per edition
Xero (JAX)	Claude-powered JAX after a March 2026 partnership with Anthropic (report); Bill Protection checks every bill before payment; a no-code agent builder	Accountant ecosystem, strong bill-fraud checks	Checks bills rather than every agent action; advice centred on bookkeeping, not capital structure
Kingdee (Lingee)	An enterprise agent operating system since May 2026 with expense-review, 24/7 bank-reconciliation, three-way-match and closing agents; AI tax and invoicing tools for small firms	Shenzhen-based leader in China's SMB and enterprise software	China-first and CAS-native; enterprise-first agents
Digits and other AI-native bookkeepers	AI accounting and bill pay for startups	Automation depth and modern design	US startups; no multi-framework support or emerging-market payment rails
Digital ledger apps and spreadsheets	Free debit-and-credit ledgers on the phone	Reach and simplicity	No analysis, no controls, no decisions

Capability	Fiona	QuickBooks	Xero	Kingdee
How agents act on money	Within an owner-set mandate	Draft for approval	Selected tasks such as payment follow-ups	Enterprise workflows
Check before execution	Independent Guardian on every action	Human approval	Bill Protection on bills	Review agents for expenses and procurement
Capital-structure advice	Core feature, stage-aware	Finance-agent insights	Cash-flow scenarios	Operational analysis
Reporting frameworks	IFRS, IFRS for SMEs, US GAAP and CAS rulepacks	Local standards per edition	Local standards per region	CAS

Fiona targets the empty corner: capital decisions with delegated authority



positioning by zone · from the competitor table above, as of 11 Oct 2026

Positions are zones, not scores: the incumbents automate the books, some acting on routine tasks; none pairs capital decisions with delegated, checked authority.

Opportunities. No one owns the position of an accountable AI participant. Emerging-market SMBs are underserved by US-centric tools. Banks need clean, explainable SMB data to lend, which makes them a distribution channel.

Threats. Xero, now built on Claude, and QuickBooks could add mandates and independent checks quickly. Kingdee owns China's SMB channel. Bank data access in emerging markets is slow to arrange.

What we do about it. Differentiate on the participation model, emerging-market fit and framework neutrality. Match incumbents only where the demo needs it, such as reconciliation and invoicing. Import from Xero, QuickBooks and Kingdee instead of fighting them, and reach small businesses through banks.

Business model and go-to-market

Fiona earns from subscriptions and reaches small businesses through the institutions that already serve them: banks, payment providers and accountants.

Channels

1. **Banks and digital lenders.** A bank offers Fiona to its small-business customers. With the owner's consent, the bank receives a credit-ready pack that makes small loans cheaper to underwrite, which suits digital banks built around inclusive finance.
2. **Accountants and bookkeepers.** One firm watches many clients from one console; Fiona does the monitoring and the accountant brings judgement.
3. **POS and payment providers.** Fiona embeds in the dashboards where sales data already lives.
4. **Direct self-serve.** A free tier brings in micro businesses, which upgrade as they grow.

Tier	For	Includes
Free	Micro businesses	Briefing, one bank account, advice only
Growth	Small businesses	All agents, agent wallet, Launch Lab, mobile approvals
Group	Medium and multi-entity firms	Roles, approval chains, consolidation, ERP connectors

Tier	For	Includes
Partner	Banks and accounting firms	Multi-client console, consented data sharing, white-label options

Price points will be set from pilot data rather than guessed now. Model costs stay low by design: high-volume work runs on the smallest model with caching and batching, and the most capable model reviews only high-value actions.

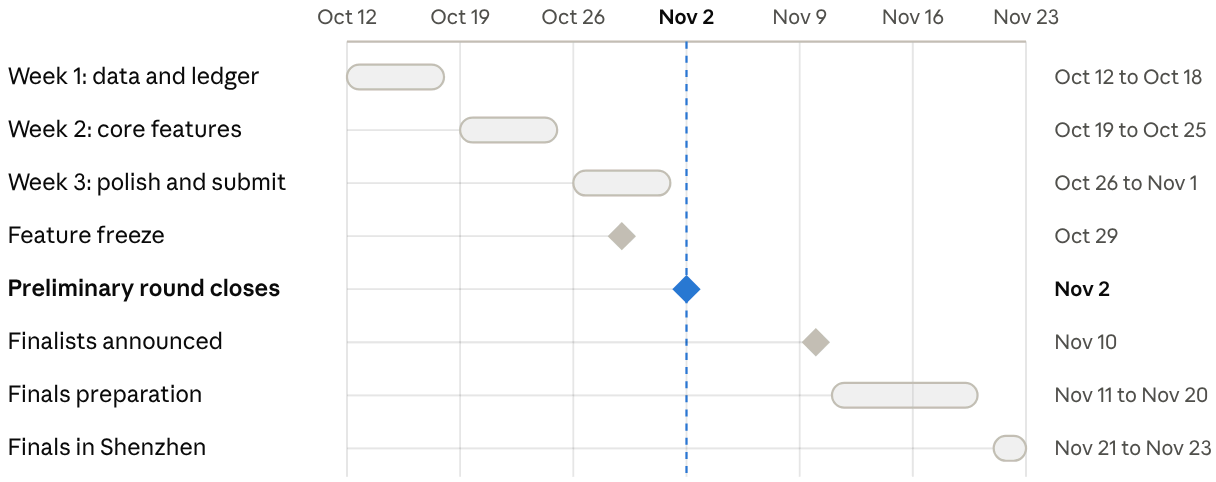
What we will measure in pilots: hours of finance work saved each month, duplicate and fraudulent payments stopped, days sales outstanding, the share of businesses that become credit-ready, and loans granted to businesses that were previously refused.

Roadmap

The preliminary round closes on 2 November 2026, finalists are announced on 10 November, and the finals run in Shenzhen on 21–23 November (IEEE CIS Shenzhen). That leaves three build weeks, so scope is cut to what the demo proves.

Three build weeks to the 2 November deadline; the finals follow 19 days later

FinTechathon 2026 International Track, weekly ticks



competition dates from IEEE CIS Shenzhen · build weeks and freeze are the team plan

The feature freeze on 29 October leaves four days for the video, the documents and rehearsals before the deadline.

Hackathon scope

Priority	What is in it
Must	Synthetic data generator, canonical ledger, bank, POS and invoice import, reconciliation and duplicate detection, the Guardian's policy, arithmetic and adversarial checks, Stage Detector, debt-versus-equity and dividend decisions, 13-week cash forecast, the morning briefing, one payment end to end through the sandbox wallet
Should	Council mode, Launch Lab, IFRS versus US GAAP view for inventory and leases, mobile approvals (Expo, or a PWA if time is short), the Guide, the second persona
Could	Urdu and Chinese voice, WhatsApp capture, XBRL import, CAS mapping, credit-ready pack
Won't, this round	Live bank or payment integrations, live tax-authority checks, audit-ready statutory statements, ERP connectors

Now, next, later

Horizon	Focus
Now, to 2 November	Everything under Must and Should, plus the video and this document
Next, November to February	Pilots with three to five real businesses, WhatsApp and photo capture, tax-authority invoice checks, bank sandbox APIs, CAS rulepack, a credit-ready pack with one lender
Later, 2027	A licensed wallet partner, ERP connectors, consolidation, the accountant console, more rulepacks reviewed by qualified accountants, security certification

Ownership

Owner	Area	Delivers by 2 November
Abdullah	Finance logic and decision engine	Decision rules and thresholds, Stage Detector specification, memo rubric, Guardian and reconciliation test cases
Asharib Khan	Architecture, interface and Claude integration	Monorepo, web app, agent runtime, Guardian service, briefing and council screens, documentation
Daniyal	Data platform and scale	Data generator and answer key, ledger schema, imports, both personas, industry packs, tenant isolation
Yahya	Forecasting and market intelligence	Cash forecast, Launch Lab, demand signals, Monte Carlo, inventory analytics

Dependencies to manage

- The data generator gates reconciliation, the Guardian and forecasting, so it is built first, in week one.
- The finance-engine API contract gates the agents and the interface; it is agreed on day two and mocked until real.
- The organisers' final submission rules (format, video length, language) shape the last week's work.

Demo script

Five minutes, one business, seven moments. Each moment proves one claim and has one presenter.

Time	Moment	What the judges see	Claim it proves	Presenter
0:00	Onboarding	Bank CSV, POS export and last year's statements go in; the Stage Detector says the owner described a growth business but the cash flows say shake-out, and shows why	Advice adapts to the business	Abdullah
0:40	Morning briefing	Three cards ranked by money at stake: a cash dip ahead, a blocked bill, slow-moving stock	Briefing first, not a chatbot	Asharib
1:20	The Guardian	The blocked bill duplicates last month's and pays a bank account changed two days ago; all four checks are shown	The AI is checked before it acts	Abdullah
2:00	Council mode	The owner asks in Urdu whether to fund a second shop with a loan or an investor; the agents debate and a memo arrives with coverage ratios and the dissent on record	Capital advice with its working shown	Asharib
2:50	Launch Lab	"Should we add a winter shawl line?" Demand signals, a forecast range, the break-even month and a cash check	Forecast before spending	Yahya
3:40	Approval by phone	A payment above the mandate reaches the phone; one fingerprint, then the ledger and audit log update live	An agent wallet with limits	Daniyal

Time	Moment	What the judges see	Claim it proves	Presenter
4:20	Persona switch	The same engine loads the Shenzhen distributor under CAS; the dashboard rebuilds and the framework lens toggles	Scale and framework neutrality	Daniyal

The closing line: *Fiona does not just answer questions about money. It takes part, within limits, and it is checked every time.*

Fallbacks. Data is pre-loaded, every moment has a recorded backup clip, and model responses for the scripted questions are cached in case the venue network fails.

Risks and mitigations

The biggest risk is not technical: it is whether an owner will let an AI near their money. The design answers that first.

Risk	Why it matters	Mitigation
Owners do not trust an AI with money	No trust, no adoption	Every agent starts at Advise; mandates are visible; every action is explained and can be stopped before it runs
A wrong number or a confident wrong recommendation	Bad decisions, lost credibility	Numbers come only from the engine; the Guardian recomputes; memos cite source rows; thin data triggers a question, not a guess
Prompt injection through bills or emails	An attacker steers a payment	Documents are data, never instructions; each agent has only the tools it needs; new or changed payees always need the owner

Risk	Why it matters	Mitigation
Messy or missing data	Weak analysis for the businesses that need it most	Photo, PDF and WhatsApp capture; model-assisted mapping; partial insights that state their gaps
Framework rules wrong or out of date	Misleading statements	Versioned rulepacks, scope limited to tested topics, accountant review before any statutory use
Payments and advice regulation	Legal exposure	Licensed partners move money; decision-support framing; compliance packs per country
Scope creep in a three-week build	A broad, shallow demo	The priority table above, a feature freeze on 29 October, and the demo script deciding what gets built
Model costs at scale	Margins shrink	The smallest capable model per task, caching, batching, and the largest model only for high-value reviews

Glossary

Term	Meaning
Agent wallet	A payment account an AI agent can use within programmable limits
Mandate	The owner's limits on what an agent may do alone
Maker-checker	A control in which one party prepares an action and another approves it
Three-way match	Checking a supplier bill against the purchase order and the goods received

Term	Meaning
Debt-service coverage ratio	Cash available for debt payments divided by the payments due
Rulepack	Fiona's versioned set of rules for one reporting framework
XBRL	A machine-readable format for financial statements, with taxonomies for IFRS and US GAAP
MSME	Micro, small and medium-sized enterprise
Z-report	A POS register's end-of-day totals

Sources

All pages accessed on 11 October 2026.

- Competition: IEEE CIS Shenzhen, International Track announcement; CLS, launch of the 8th FinTechathon
- Small-business context: IFC MSME factsheet; World Bank, SME finance; APEC, MSMEs press release
- Frameworks: IFRS Foundation, IFRS 18; Forvis Mazars, IFRS for SMEs third edition; IFRS Foundation, use of IFRS by jurisdiction; IFRS Foundation, Pakistan profile
- Methods: Dickinson (2011), Cash Flow Patterns as a Proxy for Firm Life Cycle; Amazon Science, Chronos-2
- Technology: Next.js blog; Expo SDK 56; Claude models overview
- Competitors: IT Brief, QuickBooks AI agents; Carly, QuickBooks AI tiers; Xero, Xerocon London 2026; Let's Data Science, Xero and Anthropic; Kingdee, 2026 interim results; Digits, company news4